

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

13
GREIBACH NORMAL FORM (GNF)

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REVIEW

Elimination of Left Recursion:

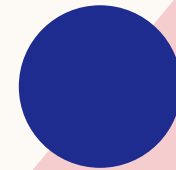
Recursive Production Rules

Elimination of Left Recursion

OUTLINE

Greibach Normal Form (GNF)

Conversion of CNF to GNF



GREIBACH NORMAL FORM (GNF)

- A CFG is in Greibach Normal Form (GNF) if the productions are in the following forms:
 - $A \rightarrow a\alpha$
where a is a terminal and α is a string of zero or more variables
- Examples
 - $S \rightarrow a \mid aAB$
 - $A \rightarrow aB$
 - $B \rightarrow cS$

GREIBACH NORMAL FORM (GNF)

- To put a CFG in GNF, it must satisfy the restrictions:
 - Already in Chomsky Normal Form (CNF)
 - No left recursion
 - Not generating ε

CONVERSION OF CNF TO GNF

- Determine an ordering for the variables or change the names of the non-terminal symbols into some A_i in ascending order of i
- Alter the rules so that the non-terminals are in ascending order, such that, if the production is of the form $A_i \rightarrow A_j \gamma$, then $i < j$ and should never be $i \geq j$
 - $i > j$: substitute
 - $i = j$: remove left recursion

CONVERSION OF CNF TO GNF

Convert the following CFG to GNF.

$(S < A < B < C < D)$

$$\begin{array}{l}
 S \rightarrow CA \xrightarrow{3} S \rightarrow aBDA \mid dBDA \\
 A \rightarrow a \mid d \\
 B \rightarrow b \\
 C \rightarrow DD \xrightarrow[1]{2} C \rightarrow aBD \mid dBD \\
 D \rightarrow AB \xrightarrow{1} D \rightarrow aB \mid dB
 \end{array}$$

- $S \rightarrow aBDA \mid dBDA$
- $A \rightarrow a \mid d$
- $B \rightarrow b$
- $C \rightarrow aBD \mid dBD$
- $D \rightarrow aB \mid dB$

CONVERSION OF CNF TO GNF

Convert the following CFG to GNF.

$(A < B < C)$

$A \rightarrow BC \xrightarrow{5} A \rightarrow bCBAC \mid aAC \mid bCBZAC \mid aZAC \mid bC$

$B \rightarrow CA \mid b \xrightarrow{4} B \rightarrow bCBA \mid aA \mid bCBZA \mid aZA \mid b$

$C \rightarrow AB \mid a \xrightarrow{1} C \rightarrow BCB \mid a \xrightarrow{2} C \rightarrow \underbrace{CACB}_{\alpha_1} \mid \underbrace{bCB}_{\beta_1} \mid \underbrace{a}_{\beta_2} \xrightarrow[3]{\text{remove left recursion}} C \rightarrow bCB \mid a \mid bCBZ \mid aZ$

$Z \rightarrow bCBACCB \mid aACCB \mid bCBZACCB \mid aZACCB \mid bCCB \xleftarrow{6} Z \rightarrow ACB$

$Z \rightarrow bCBACCBZ \mid aACCBZ \mid bCBZACCBZ \mid aZACCBZ \mid bCCBZ \xleftarrow{7} Z \rightarrow ACBZ$



PRACTICE

EXERCISES

1. Convert the following CFG to GNF.

$$S \rightarrow AS \mid a$$

$$A \rightarrow a$$

2. Convert the following CFG to GNF.

$$S \rightarrow AA \mid 0$$

$$A \rightarrow SS \mid 1$$

EXERCISES

3. Convert the following CFG to GNF.

$$\begin{aligned} S &\rightarrow a \mid AS \mid AA \\ A &\rightarrow SA \mid a \end{aligned}$$

4. Convert the following CFG to GNF.

$$S \rightarrow aSb \mid ab$$

EXERCISES

5. Convert the following CFG to GNF.

$$S \rightarrow ab \mid aS \mid SS$$

REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

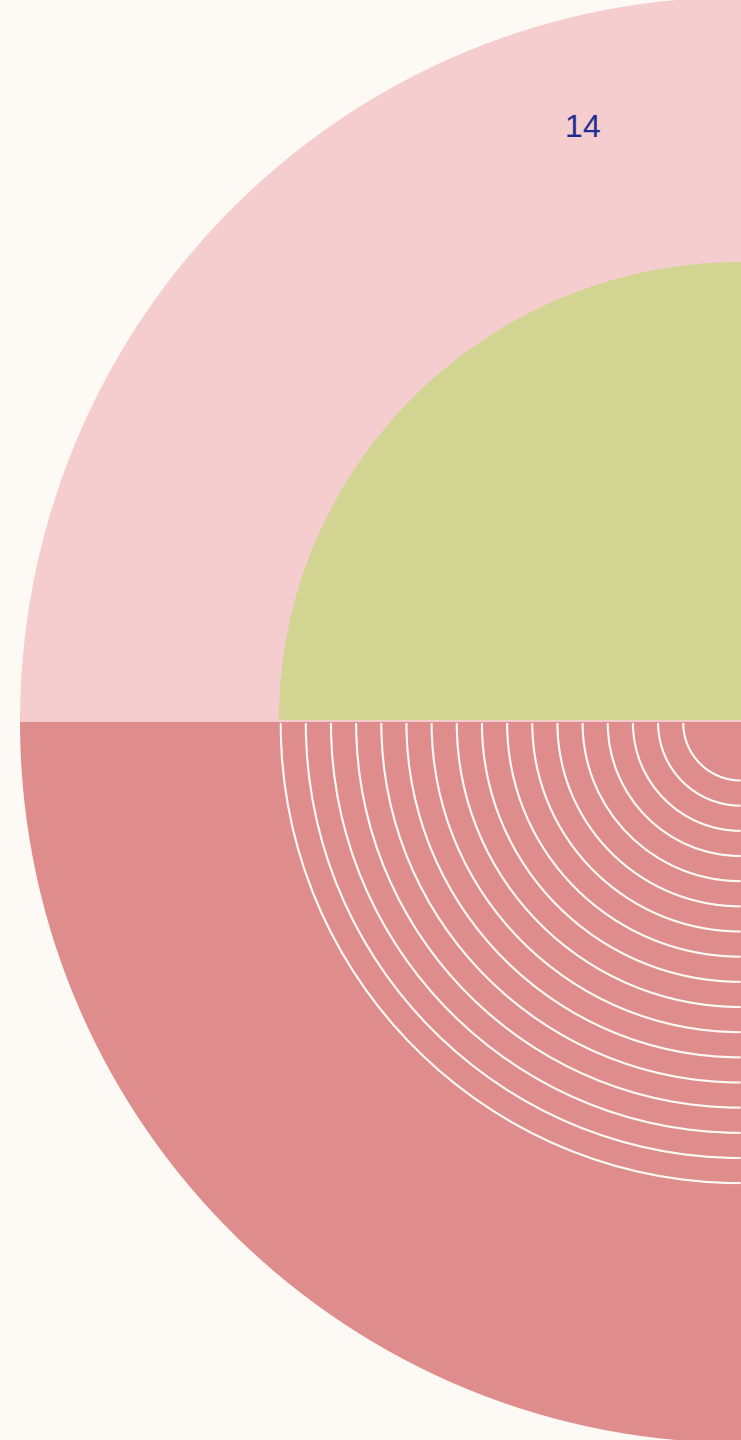
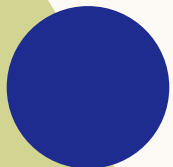
Pushdown Automata (PDA):

Pushdown Automata (PDA)

Graphical Notation for PDA

Conversion of CFG to PDA

Instantaneous Descriptions of PDA



VISION

To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

MISSION

1. To conduct studies with the best technology and curriculum, supported by professional lecturer
2. To conduct research in Informatics to promote science and technology
3. To deliver science-and-technology-based society services to implement science and technology



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